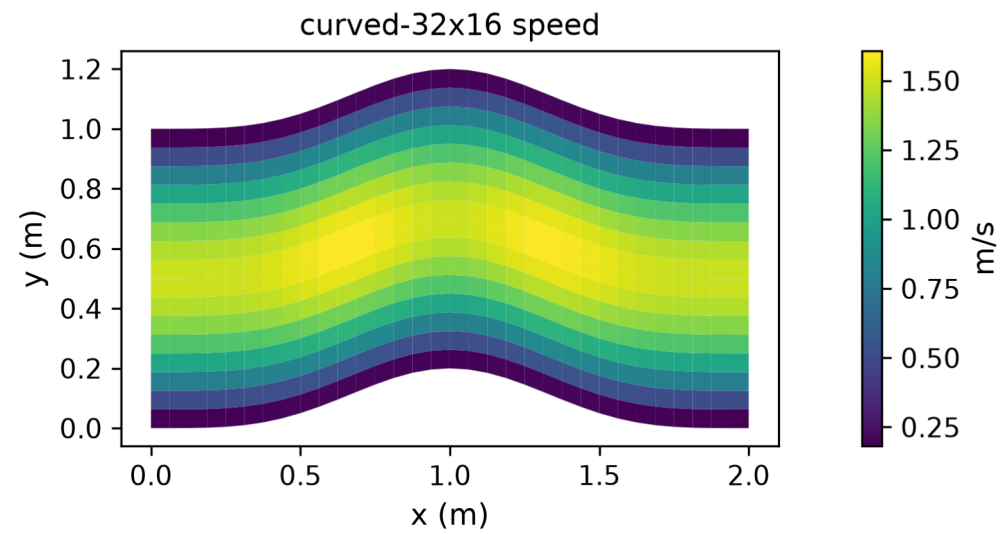
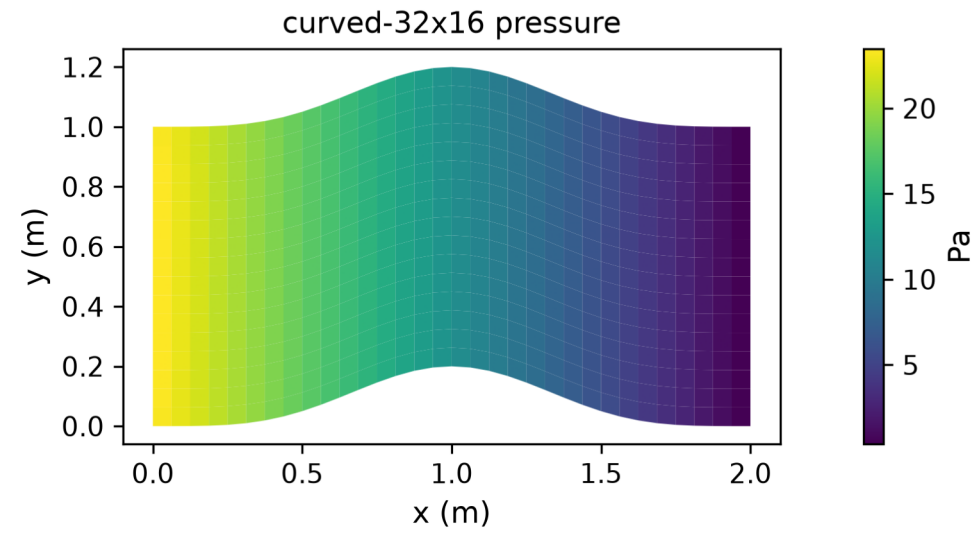
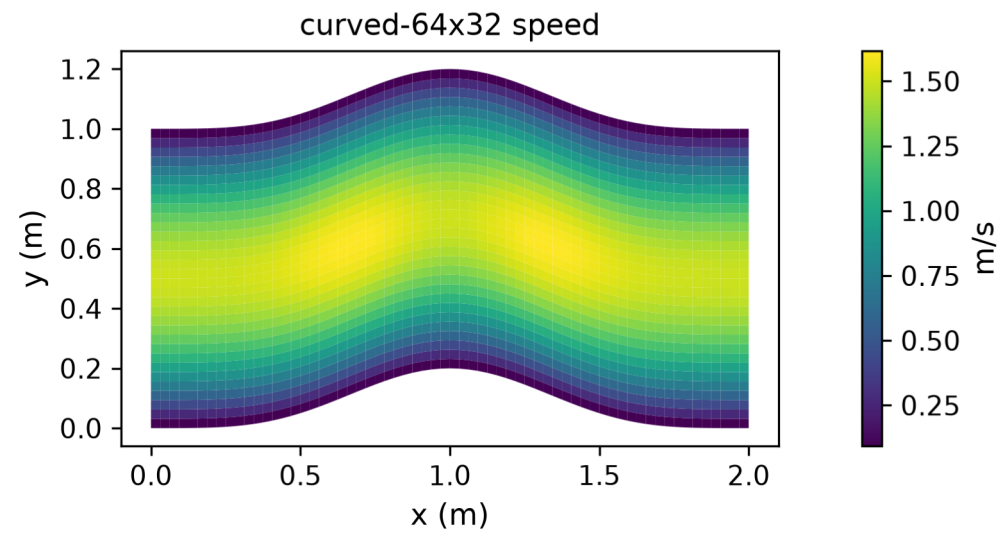
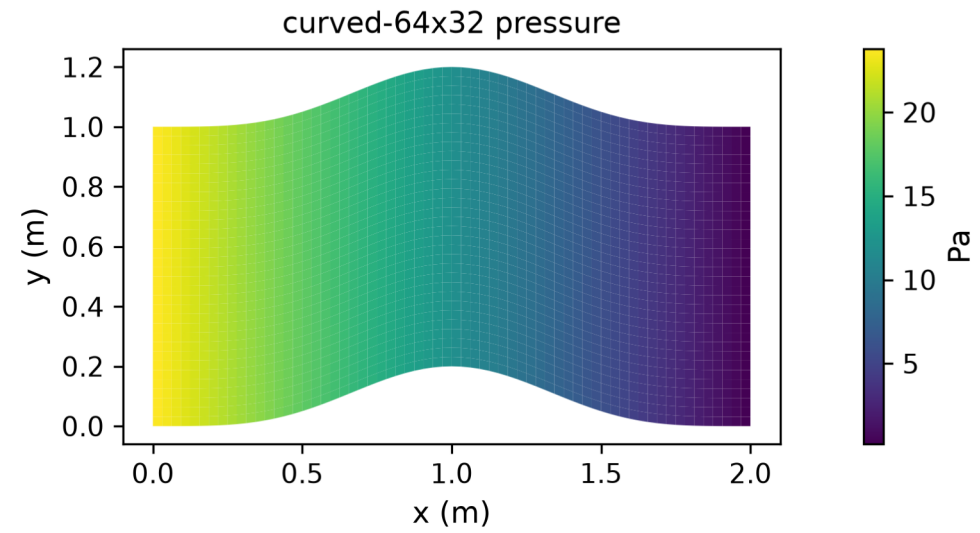


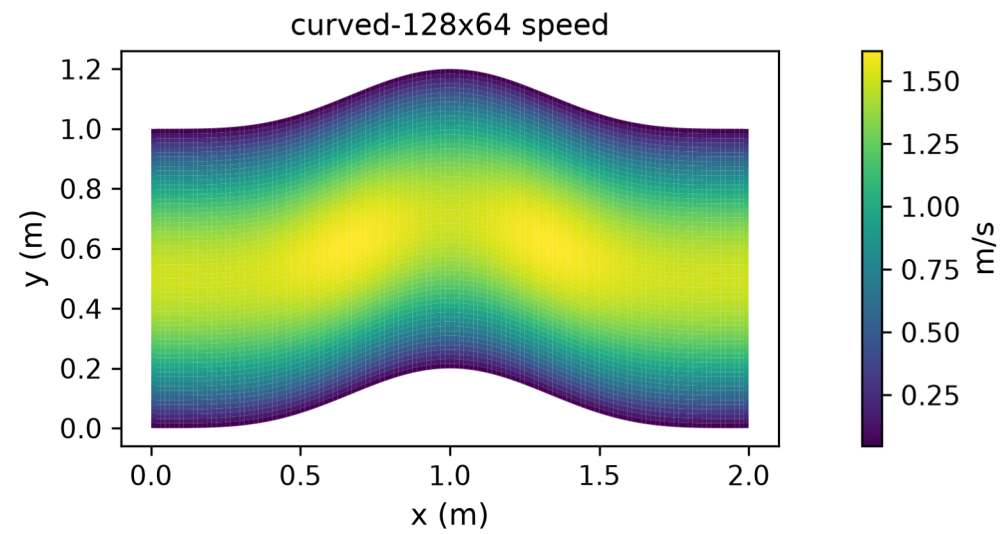
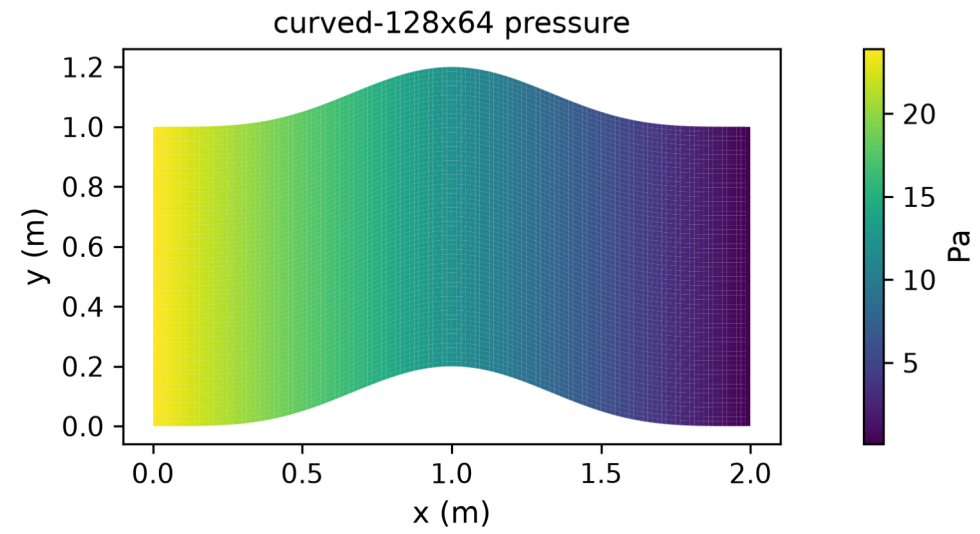
Full collocated SIMPLE: curved channel and steady SA

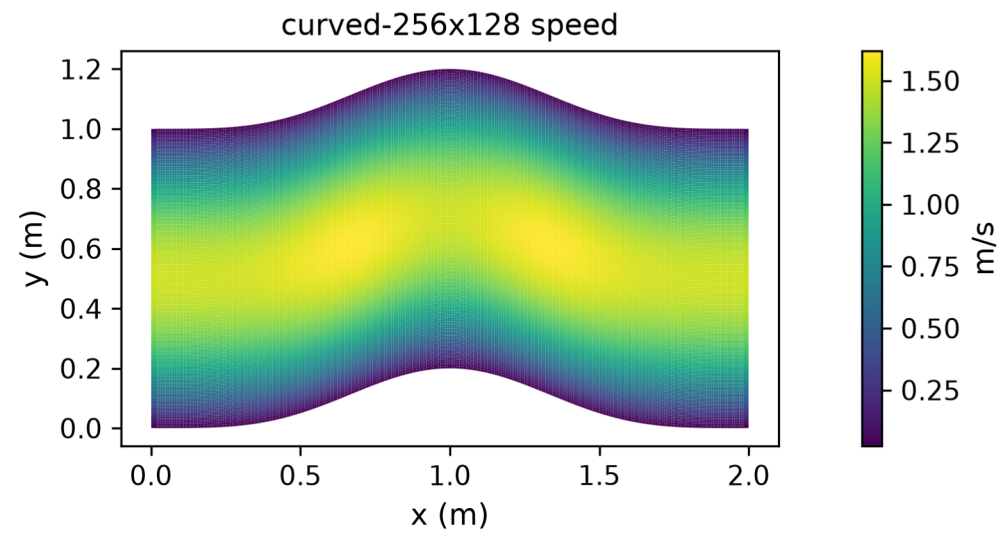
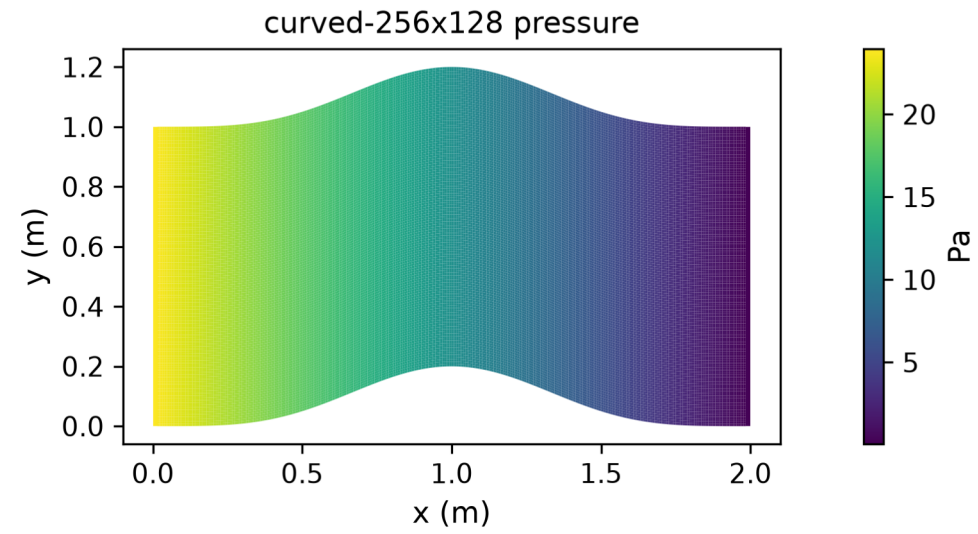
Case	CPU/CUDA steps	Converged	Physical pass	Field delta	Window CPU/GPU
curved-32x16	42/42	True	True	7.34e-15	0.678
curved-64x32	43/43	True	True	3.19e-14	1.059
curved-128x64	42/42	True	True	5.75e-14	1.985
curved-256x128	119/119	True	True	2.50e-13	3.877
sa-64x56	78/78	True	False	2.13e-13	1.269

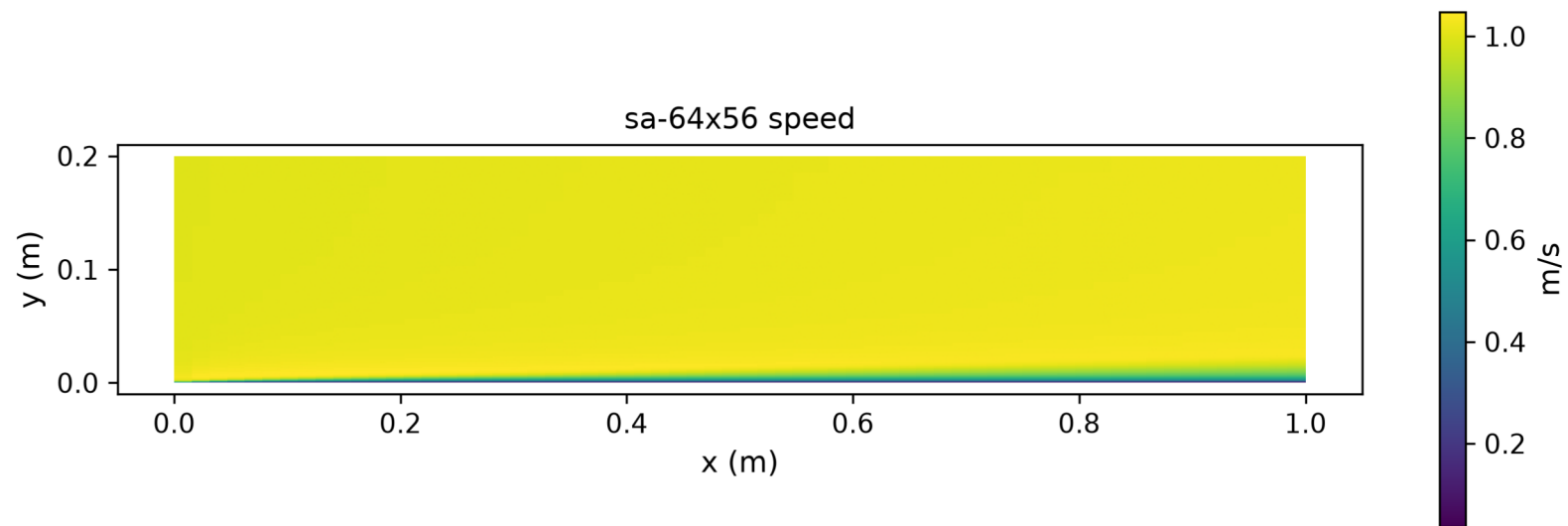
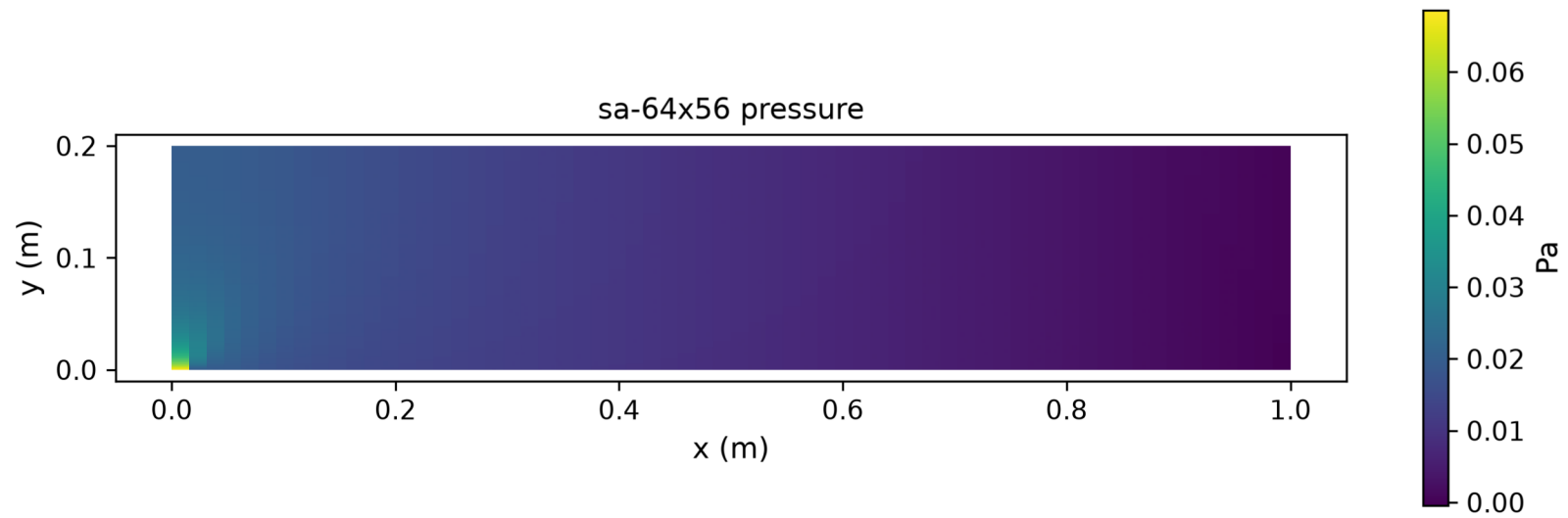
Curved continuous manufactured Navier-Stokes, with explicitly prescribed body force.
 Production SIMPLE / Rhie-Chow / nonorthogonal corrections; strict true linear residuals.
 Sparse matrices assembled on CPU; GMRES and preconditioning on actual CPU/CUDA.
 Refined initial fields interpolated from computed coarse states, not analytic data.
 Full steady solve on both devices; three repeated five-step timings are separate windows.
 SA empirical Cf gate remains independent of steady convergence and device equivalence.
 Coupled coarse correction accelerates the fine SIMPLE iteration.
 SIMPLEC / PISO / PIMPLE are not implemented or claimed.



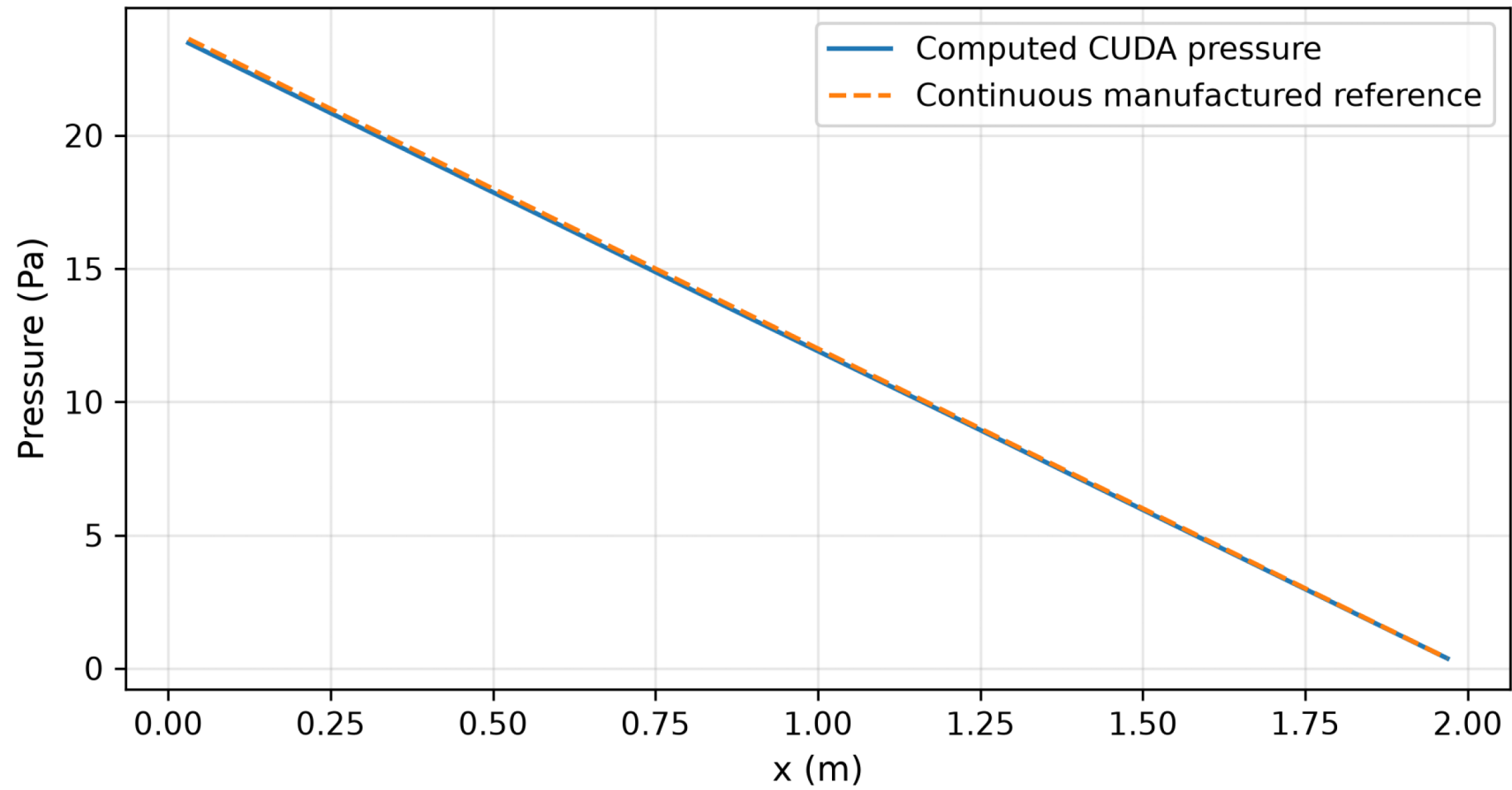


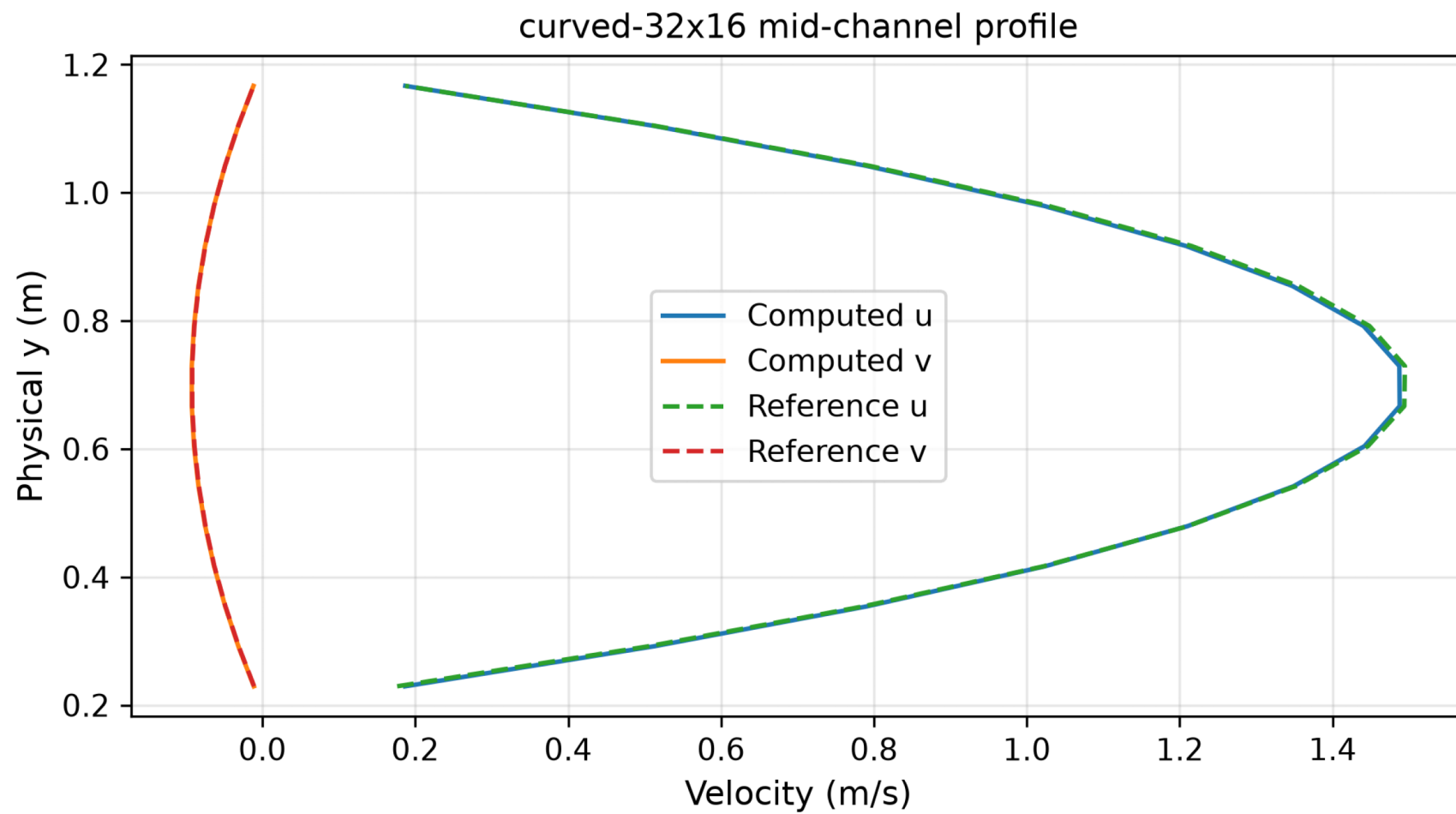




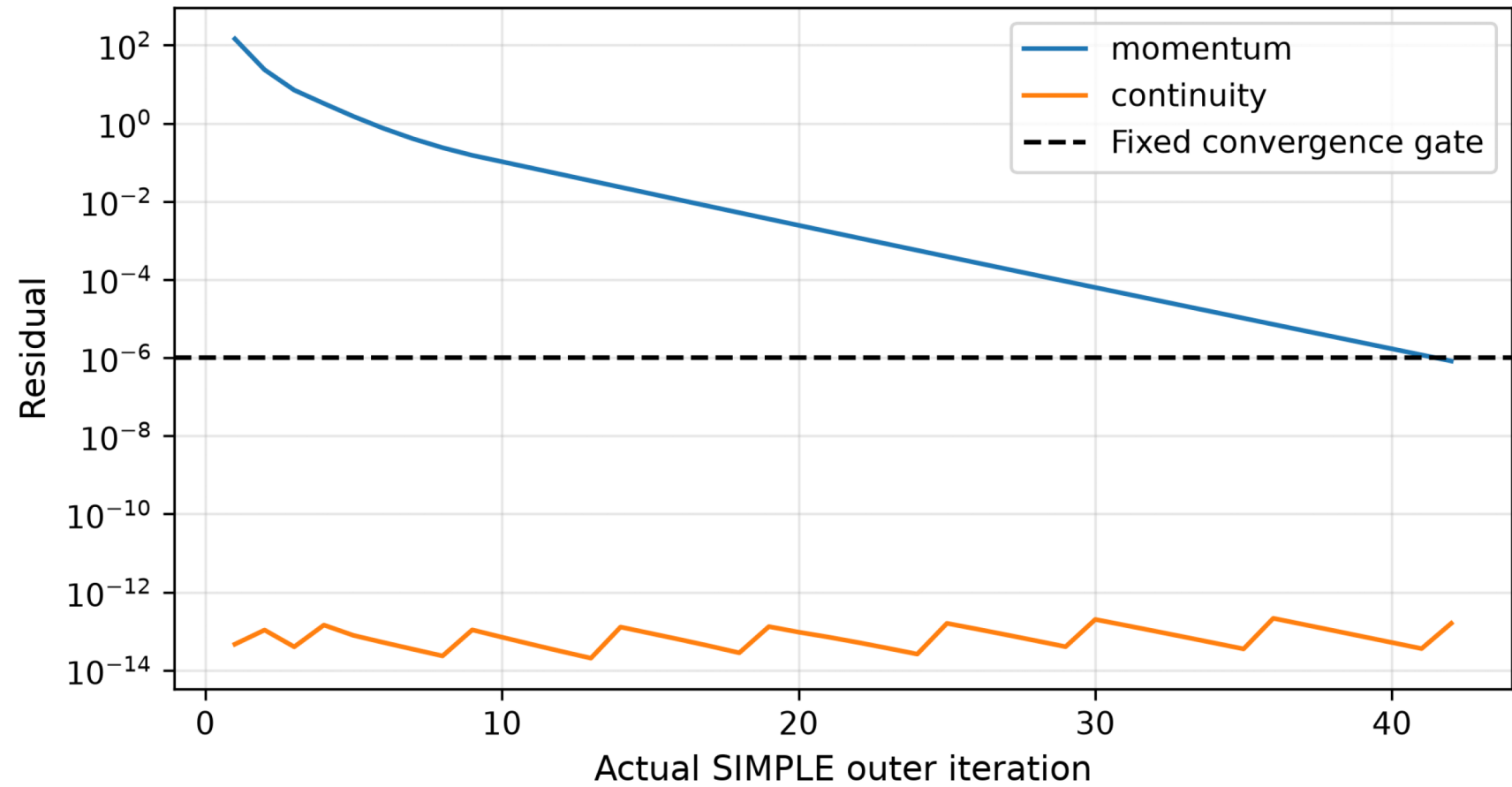


curved-32x16 center-row pressure

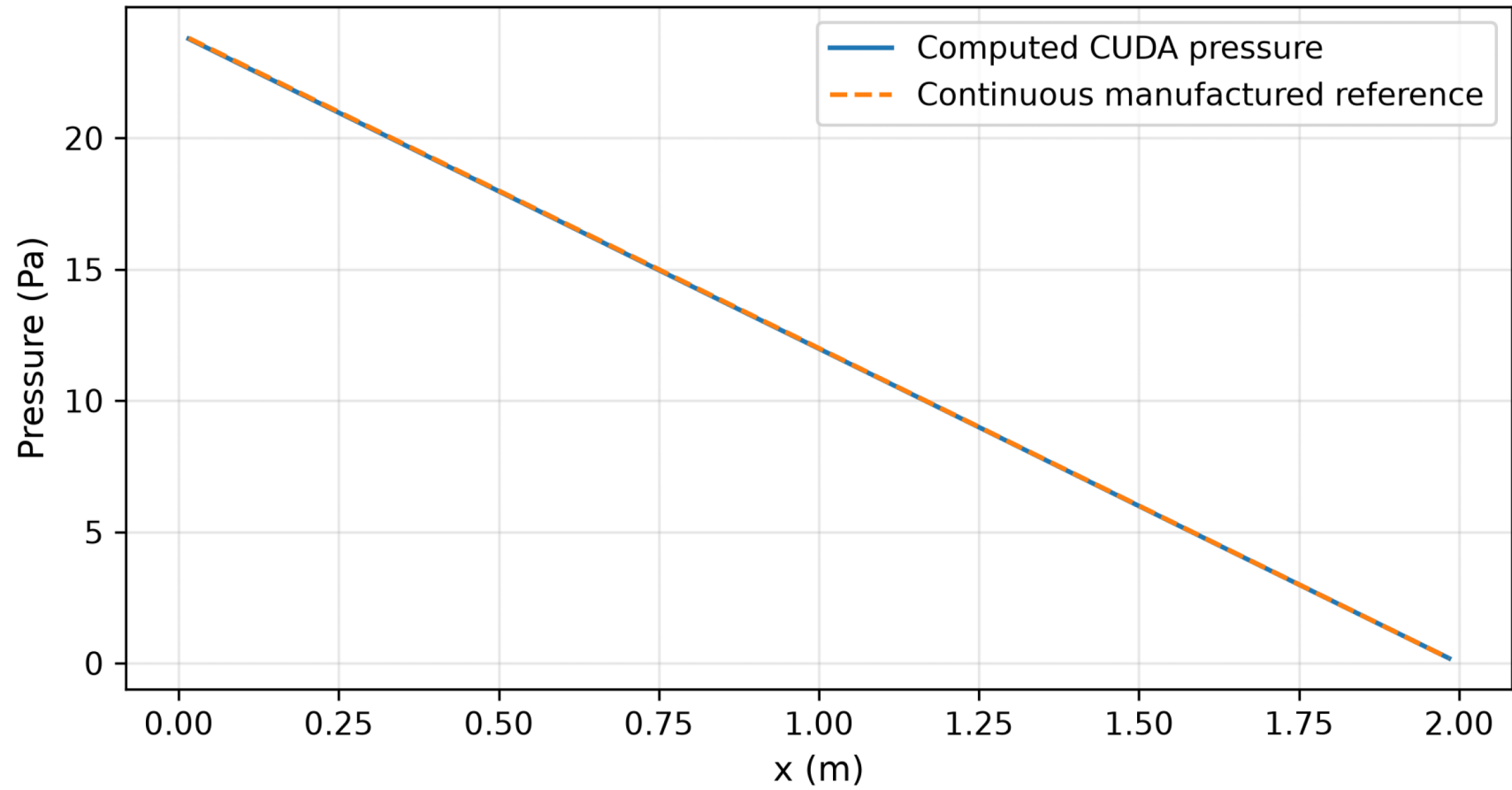




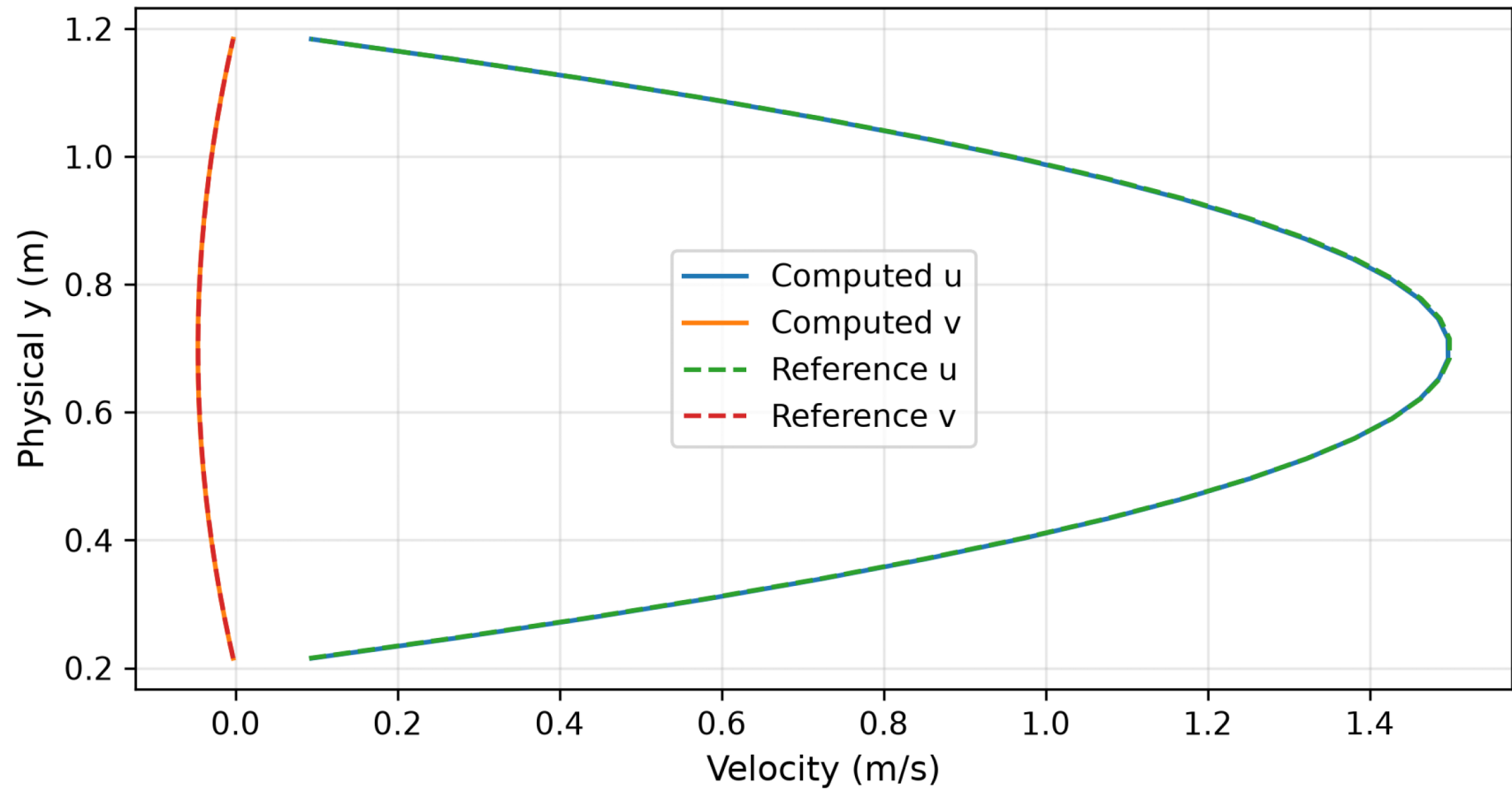
curved-32x16 convergence

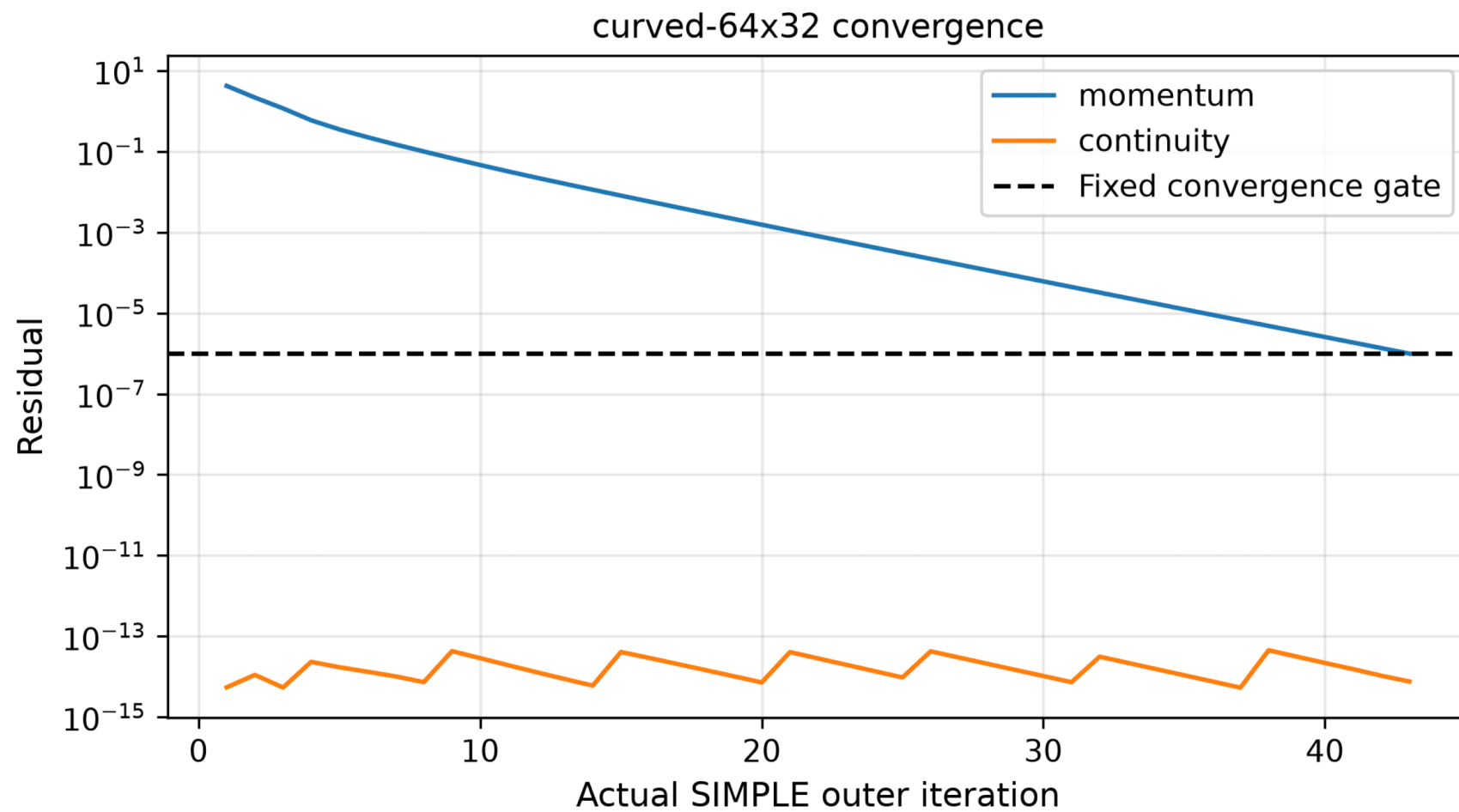


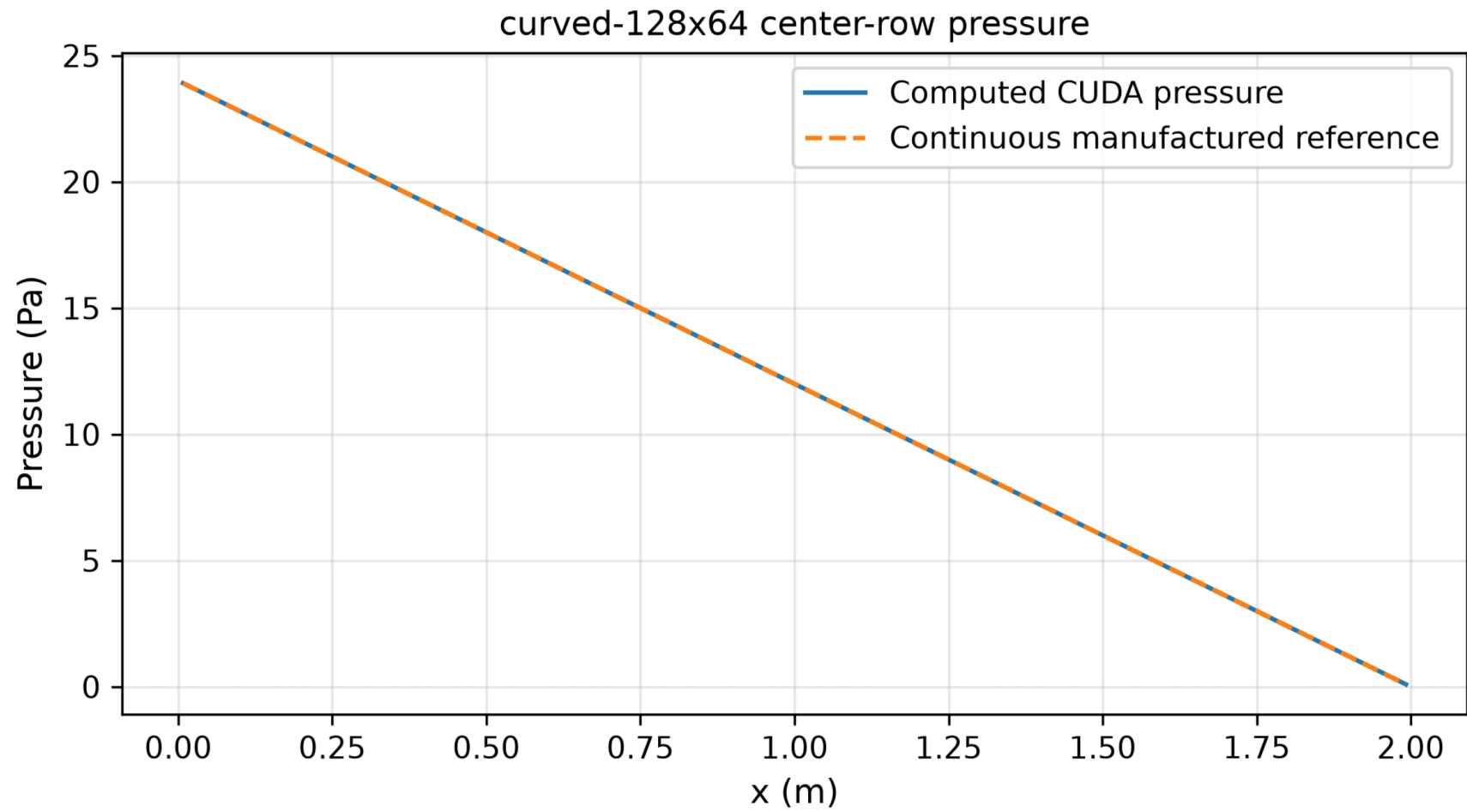
curved-64x32 center-row pressure



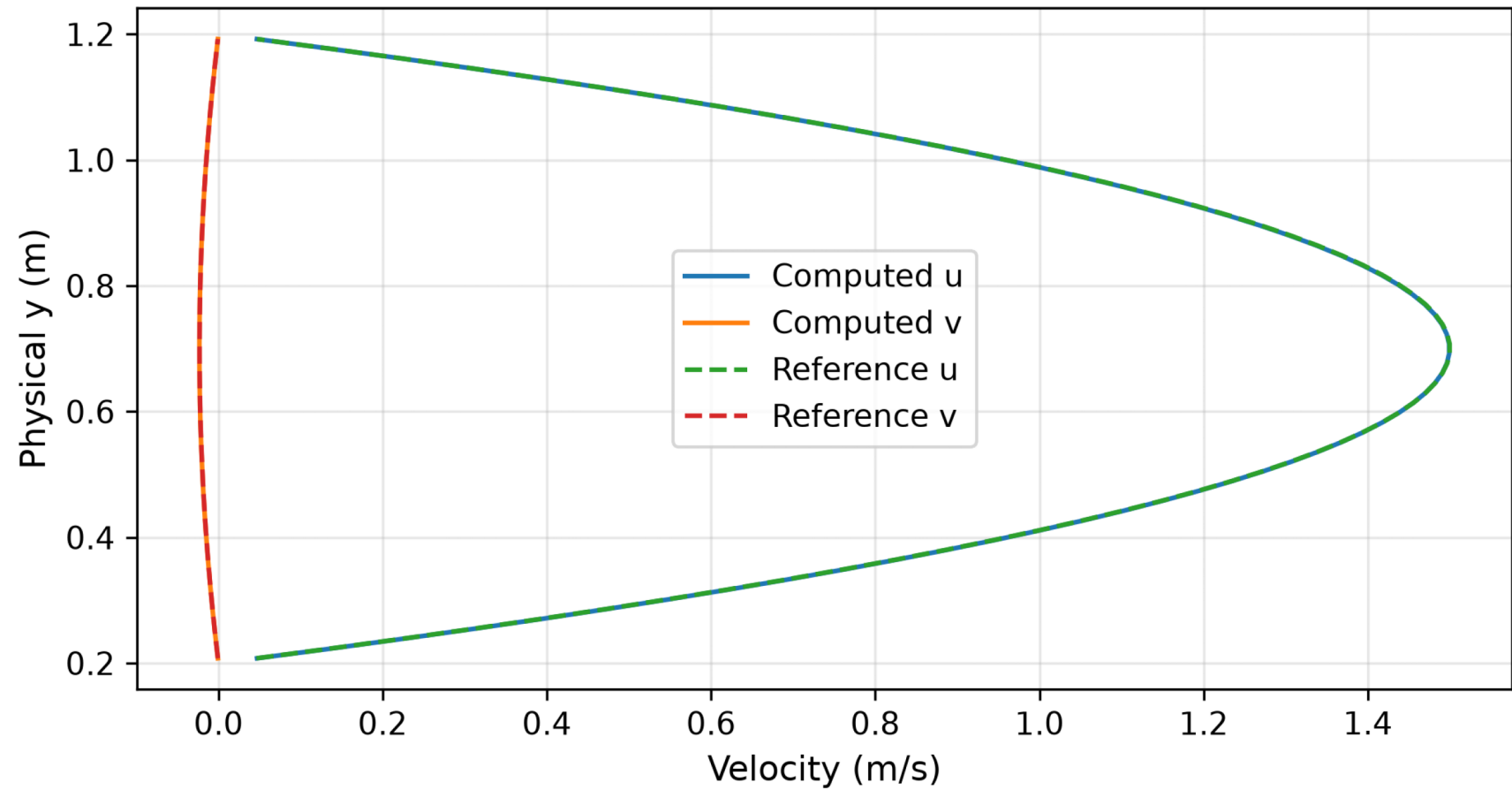
curved-64x32 mid-channel profile

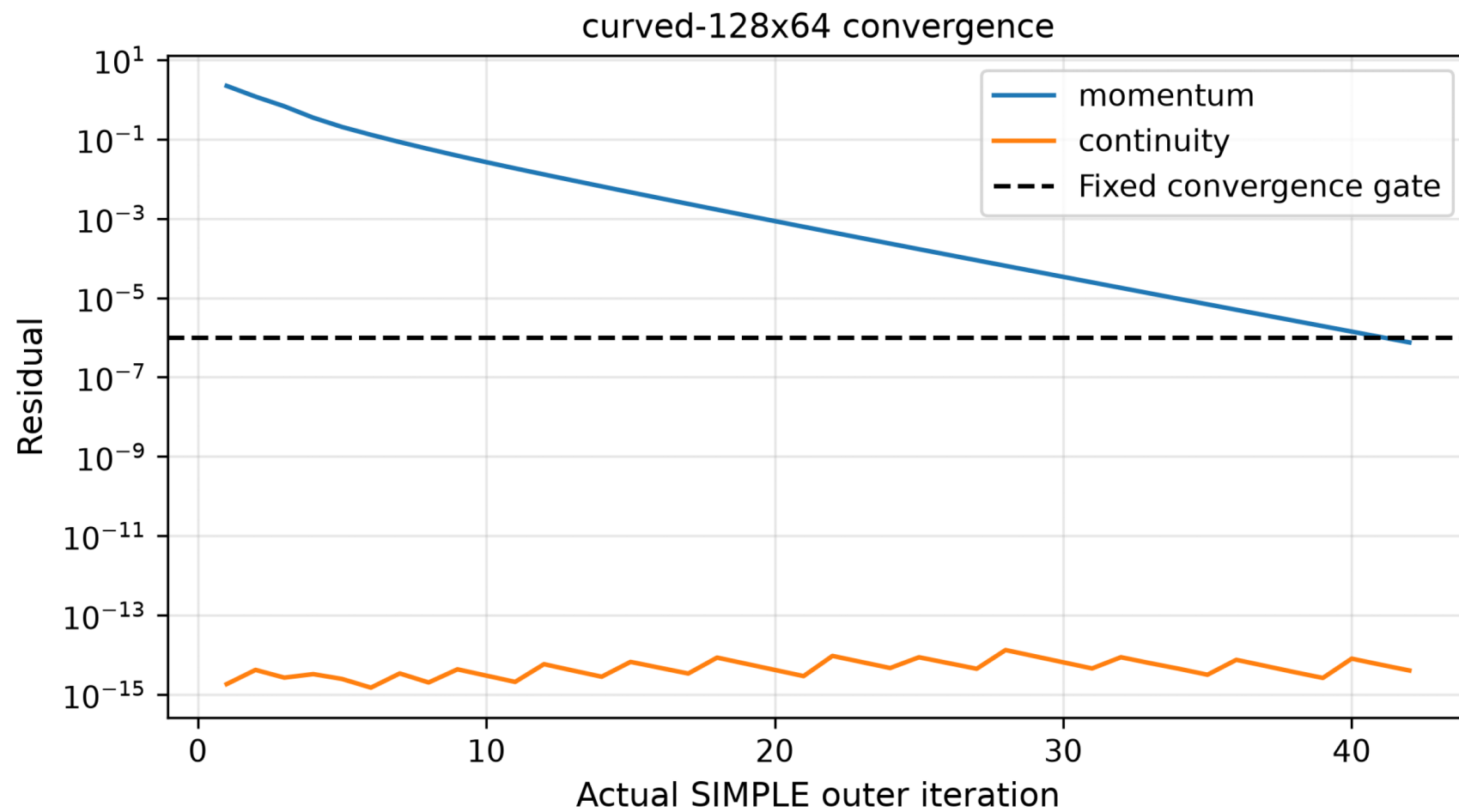


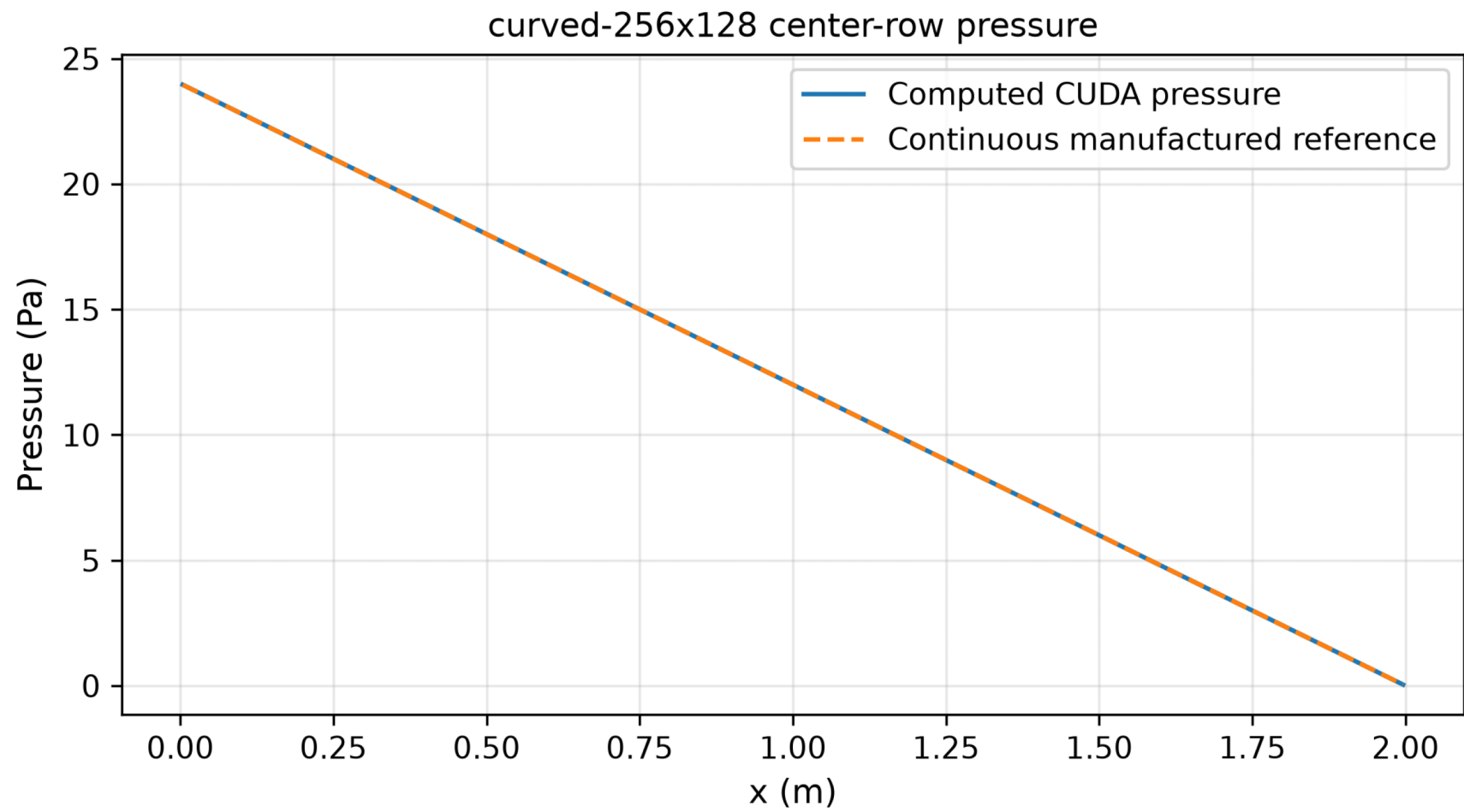




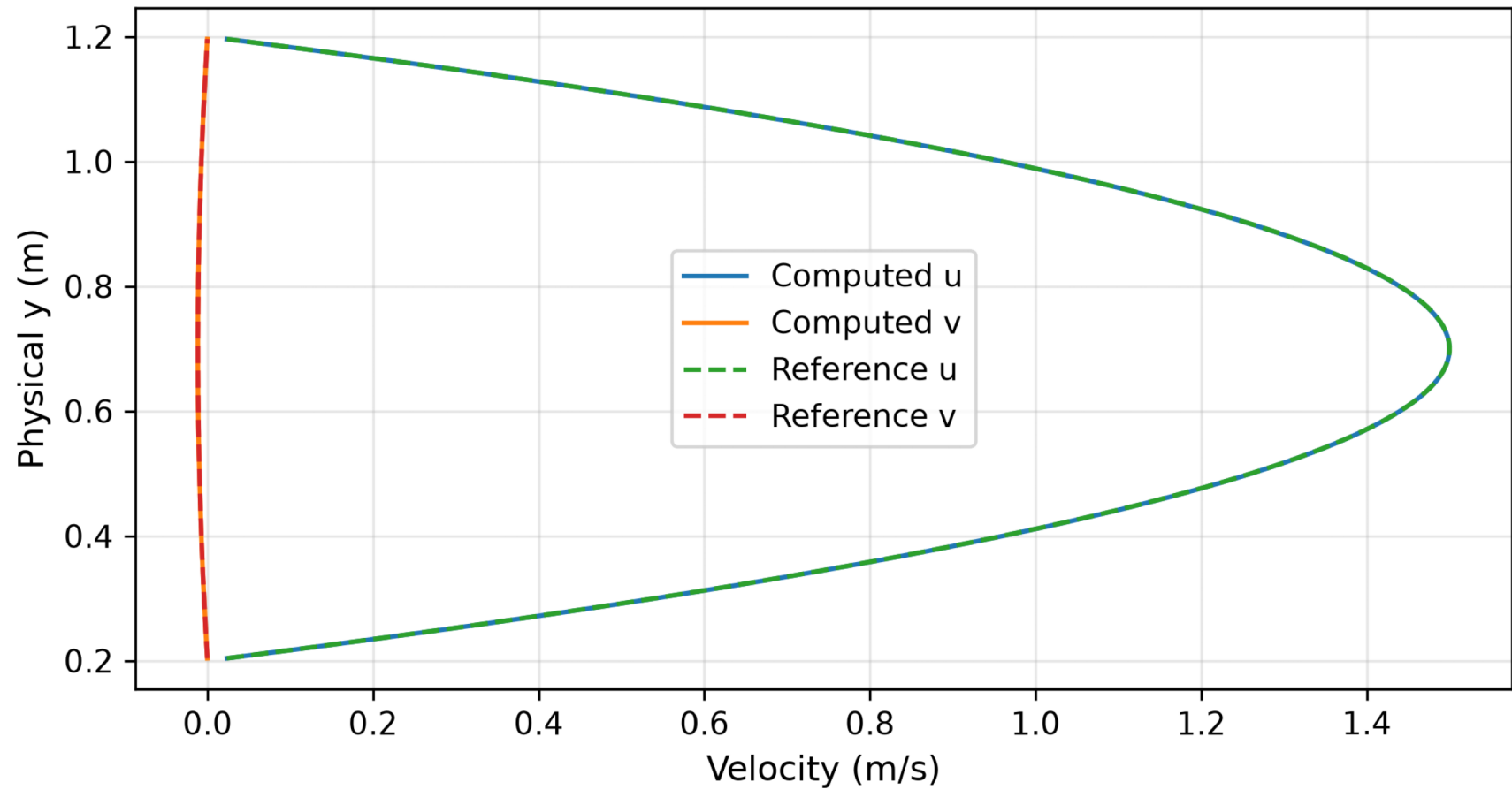
curved-128x64 mid-channel profile



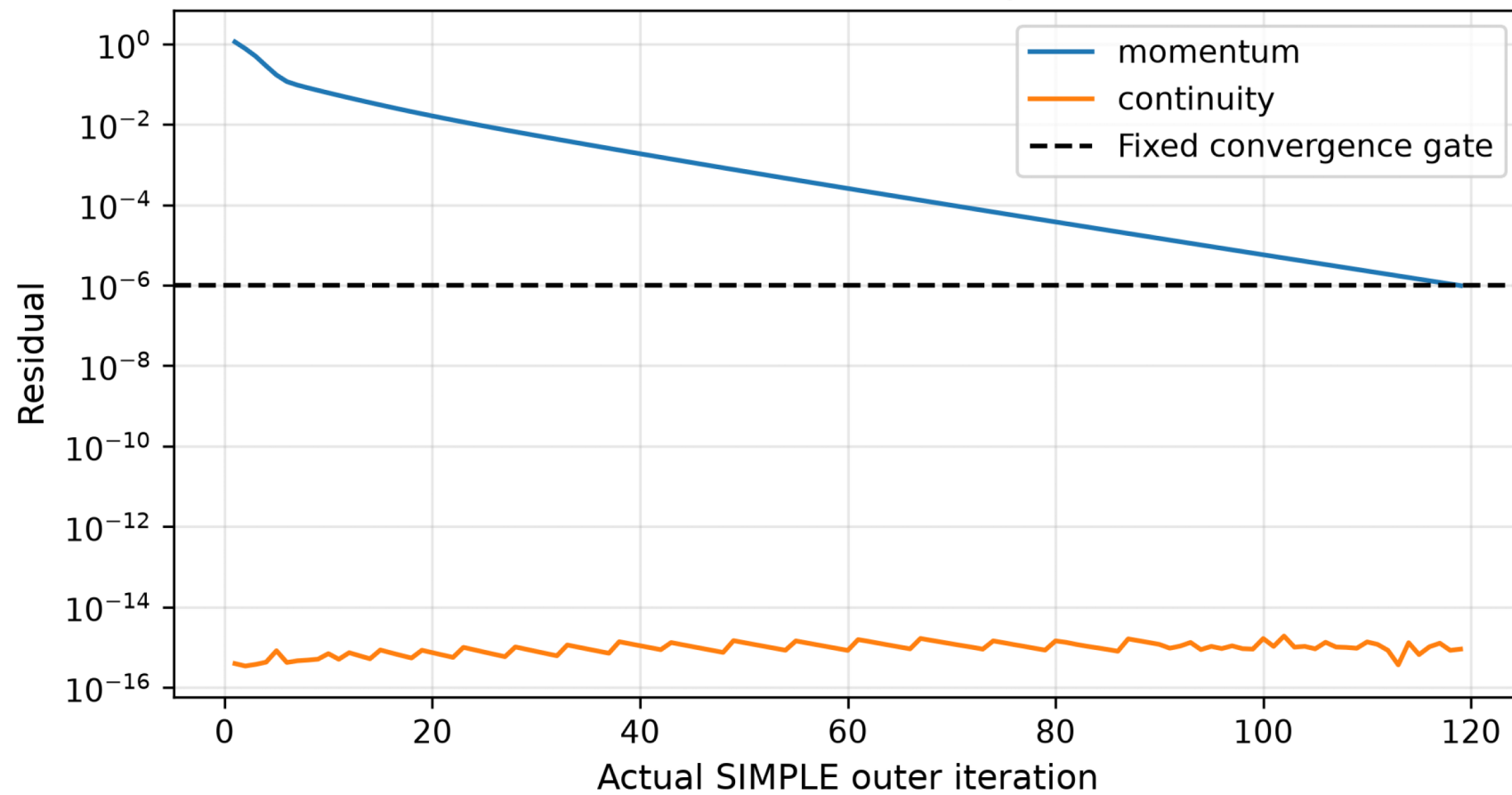


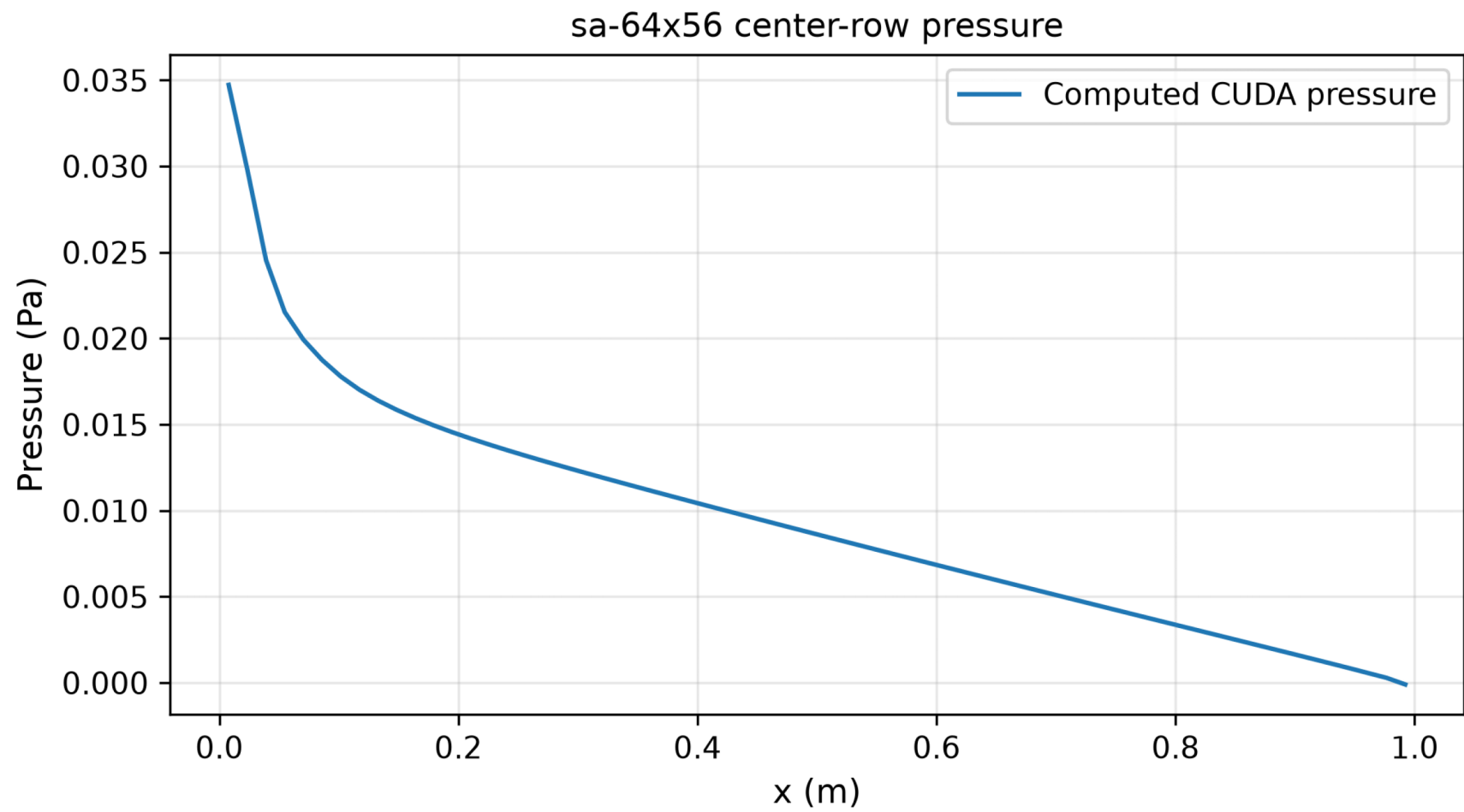


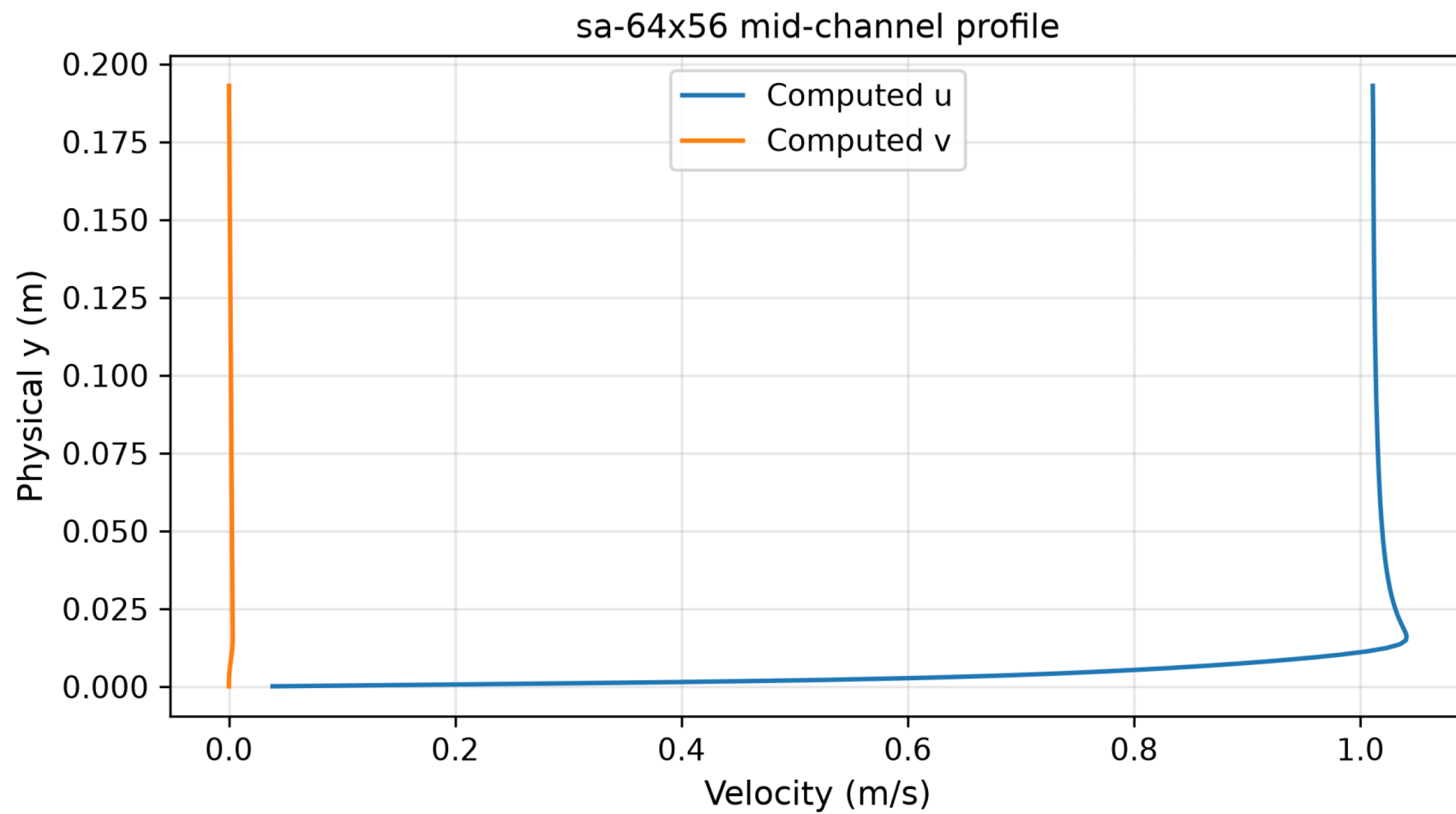
curved-256x128 mid-channel profile

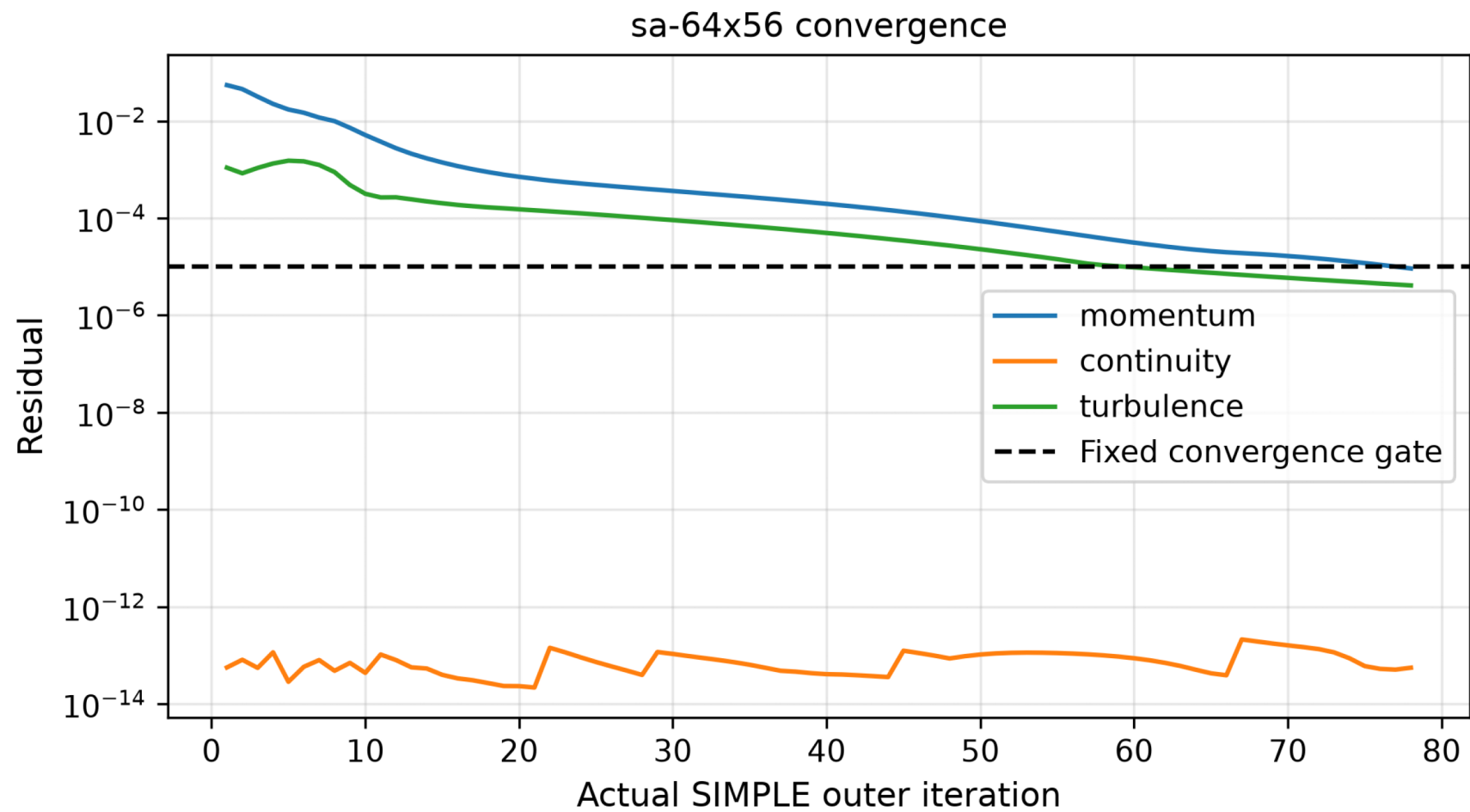


curved-256x128 convergence









sa-64x56 wall diagnostics

